

DOCUMENT 2: MATH101 Study Guide

MATH101: Calculus I — Comprehensive Study Guide & Exam Prep

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****Target Semester:**** Fall 2026

1. High-Yield Exam Topics

Based on historical performance, these topics account for over 70% of points on exams:

1. ****The Chain Rule:**** Always combined with trigonometric or exponential functions. Master nested composite functions $f(g(h(x)))$.
2. ****Optimization:**** Word problems requiring you to set up an objective function, identify constraints, find critical points, and verify max/min using the First or Second Derivative Test.
3. ****Related Rates:**** Implicit differentiation with respect to time t . Always sketch the geometric configuration first.
4. ****Integration by Parts (integral of $u \, dv = u \cdot v - \text{integral of } v \, du$):****
> ****CRITICAL EXAM NOTE:**** Integration by Parts ****ALWAYS**** appears on the Final Exam as a standalone 15-point problem or embedded in an area problem. Make sure you practice selecting u using the ****LIATE**** rule (Logarithmic, Inverse trig, Algebraic, Trigonometric, Exponential).
5. ****The Fundamental Theorem of Calculus (FTC Parts 1 & 2):**** Differentiating functions defined by integrals $d/dx \int_a^x f(t) \, dt$ and evaluating definite integrals.

Note on Textbook Coverage: ****Chapter 8 is NEVER tested.**** Do not spend study time on Chapter 8 topics (Arc Length, Surface Area of Revolution, Differential Equations applications from Ch 8). Focus strictly on Chapters 1–5 and Chapter 7.1.

2. Common Student Errors & How to Avoid Them

Limits & Continuity

* **The Error:** Plunging into L'Hopital's Rule without checking for indeterminate form $0/0$ or ∞/∞ .

* **The Fix:** Always evaluate direct substitution first. If the expression yields $5/0$, L'Hopital's Rule fails (the limit is infinite or does not exist).

Differentiation

* **The Error:** Forgetting to apply the Chain Rule when differentiating composite trigonometric expressions like $(\sin(x))^3$ or $\cos(x^2)$.

* **The Fix:** Rewrite $(\sin(x))^3$ so the outer function u^3 is clearly visible before differentiating.

Related Rates & Optimization

* **The Error:** Substituting numerical constant values for variable quantities before differentiating.

* **The Fix:** Keep quantities as variables until AFTER taking the derivative with respect to time t . Only plug in specific instantaneous numbers at the very last step.

Integration

* **The Error:** Omitting the constant of integration $+ C$ on indefinite integrals, or forgetting to change limits of integration when performing u -substitution on definite integrals.

* **The Fix:** When changing variables from x to u , immediately rewrite the lower limit a and upper limit b as $u(a)$ and $u(b)$.

3. Practice Exam Questions with Detailed Solutions

Problem 1: Limits

Question: Evaluate the limit as x approaches 0 of $(1 - \cos(3x)) / x^2$.

Solution:

1. Direct substitution yields $(1 - 1) / 0 = 0/0$ (Indeterminate). Apply L'Hopital's Rule by taking derivatives of numerator and denominator with respect to x .
2. limit as x approaches 0 of $(d/dx(1 - \cos(3x))) / (d/dx(x^2)) = \text{limit as } x \text{ approaches 0 of } (3\sin(3x)) / (2x)$
3. Substitution again yields $0/0$. Apply L'Hopital's Rule a second time:
4. limit as x approaches 0 of $(d/dx(3\sin(3x))) / (d/dx(2x)) = \text{limit as } x \text{ approaches 0 of } (9\cos(3x)) / 2$
5. Substitute $x = 0$: $(9\cos(0)) / 2 = (9 \cdot 1) / 2 = 9/2$.

Problem 2: Implicit Differentiation

Question: Find dy/dx for the curve defined by $x^3 + y^3 = 6xy$ at the point $(3, 3)$.

Solution:

1. Differentiate both sides with respect to x :
$$d/dx(x^3 + y^3) = d/dx(6xy)$$
2. Apply power and chain rule on left side, product rule on right side:
$$3x^2 + 3y^2 \cdot (dy/dx) = 6y + 6x \cdot (dy/dx)$$

3. Group terms containing dy/dx on one side:

$$3y^2 \cdot (dy/dx) - 6x \cdot (dy/dx) = 6y - 3x^2$$

4. Factor out dy/dx and solve:

$$(dy/dx) \cdot (3y^2 - 6x) = 6y - 3x^2 \Rightarrow dy/dx = (6y - 3x^2) / (3y^2 - 6x) = (2y - x^2) / (y^2 - 2x)$$

5. Evaluate at $(x, y) = (3, 3)$:

$$dy/dx \text{ at } (3, 3) = (2(3) - 3^2) / (3^2 - 2(3)) = (6 - 9) / (9 - 6) = -3 / 3 = -1.$$

Problem 3: Optimization

****Question:**** A rectangular garden is to be fenced in against a straight wall, using 100 meters of fencing. No fence is needed along the wall. Find the dimensions that maximize the enclosed area.

****Solution:****

1. Let x be the length of the two side fences perpendicular to the wall, and y be the length parallel to the wall.

2. Perimeter constraint: $2x + y = 100 \Rightarrow y = 100 - 2x$.

3. Area objective function: $A = x \cdot y = x \cdot (100 - 2x) = 100x - 2x^2$.

4. Domain constraint: $x > 0$ and $100 - 2x > 0 \Rightarrow x$ is in $(0, 50)$.

5. Find critical points by setting $A'(x) = 0$:

$$A'(x) = 100 - 4x = 0 \Rightarrow 4x = 100 \Rightarrow x = 25 \text{ meters}$$

6. Verify max using second derivative test: $A''(x) = -4 < 0$ (Concave down, so $x = 25$ is a local max).

7. Compute y : $y = 100 - 2(25) = 50$ meters.

8. ****Final Dimensions:**** 25 m perpendicular to wall, 50 m parallel to wall (Maximum Area = 1,250 square meters).

Problem 4: Definite Integration with Substitution

****Question:**** Evaluate integral from 0 to $\sqrt{\pi}$ of $x \cdot \sin(x^2) dx$.

****Solution:****

1. Choose substitution: Let $u = x^2$, then $du = 2x dx \Rightarrow x dx = (1/2) du$.

2. Change limits of integration:

* Lower limit: when $x = 0$, $u = 0^2 = 0$.

* Upper limit: when $x = \sqrt{\pi}$, $u = (\sqrt{\pi})^2 = \pi$.

3. Rewrite integral in terms of u :

integral from 0 to π of $\sin(u) \cdot (1/2) du = (1/2) \cdot (\text{integral from 0 to } \pi \text{ of } \sin(u) du)$

4. Evaluate integral:

$(1/2) \cdot [-\cos(u)] \text{ from 0 to } \pi = (1/2) \cdot (-\cos(\pi) - (-\cos(0))) = (1/2) \cdot (-(-1) + 1) = (1/2) \cdot (2) = 1.$

Problem 5: Integration by Parts

****Question:**** Evaluate integral of $x \cdot e^{(2x)} dx$.

****Solution:****

1. Choose u and dv using LIATE rule (x is Algebraic, $e^{(2x)}$ is Exponential):

* Let $u = x \Rightarrow du = dx$

* Let $dv = e^{(2x)} dx \Rightarrow v = (1/2) \cdot e^{(2x)}$

2. Apply formula integral of $u dv = u \cdot v - \text{integral of } v du$:

$$\text{integral of } x \cdot e^{(2x)} dx = x \cdot \left(\frac{1}{2}\right) \cdot e^{(2x)} - \text{integral of } \left(\frac{1}{2}\right) \cdot e^{(2x)} dx$$

3. Integrate remaining integral:

$$= \left(\frac{1}{2}\right) \cdot x \cdot e^{(2x)} - \left(\frac{1}{4}\right) \cdot e^{(2x)} + C$$

4. Factored form: $\left(\frac{1}{4}\right) \cdot e^{(2x)} \cdot (2x - 1) + C$.

4. Advice & Study Tips from Dr. Thorne

Don't skip steps: Most partial credit is lost when students skip writing out the intermediate algebraic steps.

Master trig identities: You are expected to know basic double-angle formulas ($\sin(2x) = 2 \cdot \sin(x) \cdot \cos(x)$, $\cos(2x) = (\cos(x))^2 - (\sin(x))^2$) and Pythagorean identities.

Attend Recitation: Attendance in discussion section directly correlates with a full letter-grade boost on average.